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import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report

iris = load_iris()

X = iris.data
y = iris.target

print("Iris Flower Classification using KNN")
print("-----")

print("Dataset Shape:", X.shape)
print("Classes:", iris.target_names)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

knn = KNeighborsClassifier(n_neighbors=5)

knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)

print("\nActual Values:")
print(y_test)

print("\nPredicted Values:")
print(y_pred)

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

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print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))

new_flower = np.array([[5.1, 3.5, 1.4, 0.2]])

new_flower = scaler.transform(new_flower)

prediction = knn.predict(new_flower)

print("\nNew Flower Prediction:")
print(iris.target_names[prediction[0]])

Iris Flower Classification using KNN
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*** Dataset Shape: (150, 4)
Classes: ['setosa' 'versicolor' 'virginica']

Actual Values:
[0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]

Predicted Values:
[0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 1 0 2 1 1 2 1 1 0 2 0]

Accuracy:
0.9333333333333333

Confusion Matrix:
[[10  0  0]
 [ 0 10  0]
 [ 0  2  8]]

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Classification Report:
          precision    recall  f1-score   support

   setosa          1.00      1.00      1.00         10
 versicolor    0.83      1.00      0.91         10
  virginica      1.00      0.80      0.89         10

 accuracy          0.93
 macro avg          0.94      0.93      0.93
weighted avg          0.94      0.93      0.93

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New Flower Prediction:
setosa

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